

SHUBHAM KADAM

DATA SCIENTIST | SCIENTIFIC COMPUTING & AI EVALUATION

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PROFESSIONAL SUMMARY

Data scientist and AI evaluation specialist with a B.Tech + M.Tech in Aerospace Engineering from IIT Bombay and 4+ years of freelance experience in mathematical content validation, prompt design and structured quality review. Hands-on experience in Python-based engineering optimization, statistical analysis and machine-learning systems. Combines quantitative problem-solving with clear STEM documentation and systematic assessment of model accuracy, reasoning and failure modes.

TECHNICAL EXPERTISE

Scientific programming & data analysis: Python, NumPy, Pandas, SQL and PySpark; exploratory data analysis, quantitative data processing and regex-based text processing; analysis of inventory and distribution-network datasets.

Numerical methods & engineering optimization: Foundations in numerical methods, calculus, linear algebra, differential equations, probability and statistics; Python-based engine-layout optimization.

Machine learning & statistical applications: scikit-learn, XGBoost, PyTorch and TensorFlow; classification, elasticity-based clustering and statistical analysis; evaluation of model accuracy and application-level relevance.

NLP & generative AI: LangChain applications, intent recognition, semantic similarity and enterprise search on Google Cloud Platform; open-source LLM fine-tuning with PEFT/LoRA for code-generation workflows.

AI evaluation & prompt design: Multi-step mathematics and coding prompts, adversarial testing and edge-case analysis; identification of reasoning failures and assessment of accuracy, coherence and instruction adherence.

Quality assurance & structured review: Rubric- and guideline-based evaluation, acceptance-criteria checks, error tagging and consistency review; mathematical-content validation, systematic error correction and detailed reviewer feedback.

LaTeX & scientific documentation: LaTeX and Overleaf; equation typesetting, aligned derivations, problem-solution formatting and notation consistency; preparation and verification of structured STEM documentation.

Data annotation & transcription: Air traffic control transcription, ICAO phonetics and voice activity detection (VAD); speech-segment timestamps, speaker-turn labels and systematic quality review of audio datasets.

PROFESSIONAL EXPERIENCE

Data Scientist II | K India

May 2024 - Present

Bengaluru, India

- Optimized pricing grids for a GCC bank using elasticity-based clustering, delivering approximately SAR 100 million in incremental revenue.
- Built GenAI and NLP systems for intent recognition and routing optimization, achieving 96% classification accuracy.
- Developed a LangChain copilot for CAPA and deviation analysis, reducing analysis time by approximately 90% with 96% relevance.

Data Scientist | KPMG India

Jul 2022 - May 2024

Bengaluru, India

- Built a Python minimum viable product for engine layout optimization, reducing design cycles from approximately two months to 30-40 minutes.
- Analyzed FMCG distribution-network data to identify productivity variability and support digital-twin decision-making.
- Fine-tuned open-source large language models using PEFT/LoRA, automating approximately 25-50% of code generation for legacy systems.

Supply Chain Analyst Intern | Tata Steel

Jun 2021 - Jul 2021

India

- Used SQL and Python to analyze five years of data for 28,000+ spare parts, proposing inventory value reduction strategies totaling INR 360 million.
- Identified tail-spend opportunities and recommended vendor contract changes with projected savings of 9%.

PROFESSIONAL EXPERIENCE (CONTINUED)

Summer Intern | Deloitte

Jul 2020 - Aug 2020

India

- Built an NLP and semantic-similarity enterprise search application on Google Cloud Platform, achieving approximately 3 ms search times.

AI EVALUATION & STEM CONTENT EXPERIENCE

AI Prompt Engineer | Outlier AI

Remote | Freelance | Part-time

- Designed complex, multi-step prompts in mathematics, coding and domain-specific reasoning to evaluate AI model capabilities.
- Stress-tested model responses for edge cases, logical inconsistencies and reasoning failures; assessed accuracy, coherence and instruction adherence.
- Documented model weaknesses and provided detailed feedback to support AI evaluation, safety and alignment improvements.

Researcher | Prolific

Sep 2021 - Present

Remote | Freelance | Part-time

- Prepared and verified LaTeX problem-solution pairs from STEM educational materials, checking mathematical expressions, notation consistency and formatting.
- Evaluated the authenticity and quality of prompts and AI responses using structured review methods.
- Performed guideline-based quality assurance across 100+ tasks, logging errors and correcting inconsistencies.

AI Trainer | Invisible Technologies

May 2021 - Present

Remote | Freelance | Part-time

- Produced and reviewed LaTeX equations, aligned derivations and mathematical content for correctness, consistent notation and formatting standards.
- Validated AI outputs against task-specific acceptance criteria, including edge cases, mathematical accuracy and instruction-following compliance.

ATC Transcription Specialist | Alignerr

Remote | Freelance | Part-time

- Transcribed air traffic control recordings using domain-specific terminology and ICAO phonetics, maintaining strict transcription standards.
- Performed voice activity detection (VAD) annotation with speech segments, timestamps and speaker-turn labels; reviewed audio batches for errors and consistency.

EDUCATION & RESEARCH

B.Tech + M.Tech, Aerospace Engineering

2017 - 2022

Indian Institute of Technology (IIT) Bombay | Grade: O (Outstanding)

Publication: Physical Design aware Scan Cell reordering for Low power testing (EWDTS 2022).

Relevant coursework: Numerical Methods, Optimization, Differential Equations, Linear Algebra, Probability & Statistics, Calculus.

Class XII (CBSE), PCM | Jawahar Navodaya Vidyalaya (JNV) | 2016 - 2017 | Grade: A

CERTIFICATIONS

Machine Learning Specialization - Stanford University / Coursera (Andrew Ng)

Deep Learning Specialization - DeepLearning.AI / Coursera

Natural Language Processing Specialization - DeepLearning.AI / Coursera

LangChain for LLM Application Development - DeepLearning.AI

Python for Data Science and Machine Learning - Udemy

LANGUAGES

English (C1, Advanced) | Hindi (C2, Native/Bilingual) | Gujarati (C2, Native/Bilingual)